

■ Dt

$\text{Dt}[f, x]$ gives the total derivative $\frac{d}{dx} f$.

$\text{Dt}[f]$ gives the total differential df .

$\text{Dt}[f, \{x, n\}]$ gives the multiple derivative $\frac{d^n}{dx^n} f$.

$\text{Dt}[f, x_1, x_2, \dots]$ gives $\frac{d}{dx_1} \frac{d}{dx_2} \dots f$.

$\text{Dt}[f, x_1, \dots, \text{Constants} \rightarrow \{c_1, \dots\}]$ specifies that the c_i are constants, which have zero total derivative. ■ Symbols with attribute `Constant` are taken to be constants, with zero total derivative. ■ If an object is specified to be a constant, then all functions with that object as a head are also taken to be constants. ■ All quantities not explicitly specified as constants are assumed to depend on the x_i . ■ Example: $\text{Dt}[x y] \longrightarrow y \text{Dt}[x] + x \text{Dt}[y]$.

■ $\text{Dt}[x y, \text{Constants} \rightarrow \{x\}] \longrightarrow x \text{Dt}[y]$. ■ You can specify total derivatives by assigning values to $\text{Dt}[f]$ etc.

■ See page 413. ■ See also: `D`, `Derivative`.